S. No.	Course Category	Total Credits
A	**Basic Science and Engineering Courses (BSEC) (33 Credits)**	
	BSEC : 15 Courses | 33 Credits	
	Foundation Courses are necessary to study further engineering courses. These include, Mathematics, Physics, Chemistry and Humanities.	**33**

B	**Professional Core Courses (PCC) (50-60 Credits)**	
	PCC : 18 Courses | (50 – 60 Credits)	
	Provides in-depth knowledge of core engineering concept through theory and hands-on learning. These courses prepare the students for careers relevant to their field of study and higher studies.	**56**
1	**Electrical Circuits, Electronics & Instrumentation domain**	**19**
2	**Electrical Machines, Power Electronics & Drives domain**	**18**
3	**Power Systems & Control Engineering domain**	**19**

C	**Inter Disciplinary Track (Emerging Technology Vertical)** **IDT (EmTV) (Cr:09)**	
	IDT (EmTV) : 3 Courses | 09 Credits	
	Provides exposure to interdisciplinary and emerging technology domains beyond core engineering. Encourages innovation, adaptability, and readiness for future technology-driven careers.	**09**
	Track - A Emerging Technology Courses Track (3 Courses, Each 3 Credits)	

Offering Dept.	EmTV (3 Course, Each 3 Credit)	Who can study?
ECE / EEE	Sensors & IoT	CSE, CIVIL, IT, CSBS and AI&DS
MECH	Renewable Energy System	ECE, CSE, CIVIL, IT CSBS and AI&DS
CSE	Front End Technologies	MECH, ECE, EEE and CIVIL
CSBS	CRM Fundamentals	MECH, ECE, EEE and CIVIL

Track - B
Open Ended Courses (Each 3 Credits)
No. of. Open Ended Courses: 17

Offering Dept.	Open Ended Course	Offering Dept.	Open Ended Course
MECH	Sustainability Engineering	CSBS	Mobile App Development
MECH	Occupational Health and Safety	CSBS	No Code Application Development
ECE	Consumer Electronics	CSBS	Low Code Application Development
ECE	Wearable Devices	AI&DS	Data Analytics
EEE	Drone Technology	CDT	Japanese Language –Level 1
CIVIL	Smart Buildings	CDT	Japanese Language –Level 2
CSE	Data Visualization	CDT	Japanese Language –Level 3
CSE	Robotics Process Automation	iStart	Design Thinking
CSE	Digital Marketing	iStart	Start-up idea to market
IT	Augmented Reality / Virtual Reality		

D	Industry Ready Track / Department Industry Vertical IRT (DIV) (18 Credits)	
	IRT (DIV) : 6 Courses | 18 Credits	
	Professional Elective courses are offered as Department Industry Verticals (DIVs). Each DIV has 6 courses on specific technology domain related to the branch of study. Industry experts design and teach these courses. Each DIV makes the students industry ready or ready to deploy in the given domain. Students can choose any one DIV.	**18**
	List of DIVs	

DIV	Courses
Power Electronics and Electric vehicles	• Electrical Vehicle Architecture and Powertrain System • Power Converters and Motor Control • EV Charging Technology • Electrical Vehicle Design • Energy Storage and BMS • Autonomous and Connected Vehicles • Functional Safety and Reliability in EV • Cybersecurity in Connected Vehicles
Automotive Electronics	• Embedded Hardware Design • Embedded Programming for Automotive • Automotive ECU Development • Testing and Simulations of Automotive ECUUs • Model-Based Application ECUs Logic Development • AUTOSAR-Based ECUs Development
Embedded Systems and IoT	• Embedded C Programming • ARM Processors • Embedded System Interfacing • Linux Operating System • Real Time Operating System • TCP/IP for Embedded Systems

E	Digital & Artificial Intelligence Track (DAIT) (26 Credits)	
	DAIT : 7 Courses | 26 Credits	
	Equips students with future-ready digital competencies. Develops strong programming, analytics, AI, and IoT skills for modern engineering applications. Opens pathways to digital careers, smart technologies, and interdisciplinary innovation.	**26**

F	Innovation and Creativity Track (ICT) (15 Credits)	
	ICT: 5 Courses | 15 Credits	
	To prepare the students on Innovation, Complex Problem Solving, Creativity, Teamwork & Collaboration, Critical Thinking and learning beyond content, ICT is introduced.	**15**

G	Employability Enhancement & Life Skills Track (EE & LST) (13 Credits)	
	EE & LST : 6 Courses | 13 Credits	
	To develop English Communication Skills, Group Discussion Skills, Aptitude and Problem Solving Skills and Interview Skills. To develop Life Skills, Soft Skills, Universal Human Values and Strong Professional Ethics.	**13**

H	Teaching - Learning Practices with Higher Order Thinking Skills TLP (HOTS)	
	TLP (HOTS) : All Courses | Sem I to VII	
	Today industry require capability of fresh recruits to solve real life complex problem at Higher Order Thinking levels (Analyze, Evaluate and Create) in the blooms taxonomy of learning. The TLP of all courses are redesigned in such a way that students solve "create level" problems in addition to regular textbook "apply level" problem solving.	

I	Honours / Minor Degree Options	
	Honours/Minor: 6 Courses | 18 Credits	
	Honours and Minor programmes enable students to gain advanced and interdisciplinary skills beyond the regular curriculum, enhancing employability, innovation, and industry readiness.	
	(see table below)	**18**

Honours	Any one of the DIVs can be chosen in own department
Minor Specialization	**Courses**
Artificial Intelligence and Machine Learning	✓ Fundamentals of Artificial Intelligence ✓ Introduction to Machine Learning ✓ Deep Learning and Neural Networks ✓ Introduction to Data Analytics ✓ Fundamentals of Data Warehousing and Mining ✓ Ethics in AI
Data Analytics	✓ Data Science Fundamentals ✓ AI & ML Fundamentals ✓ Exploratory Data Analytics ✓ Visualization Techniques ✓ Fundamentals of Data Warehousing and Mining ✓ Ethics in AI